

Phishing URL Detection System

Project Report — Spoorthy Madduri
Machine Learning | NLP | REST API | Explainable AI

1. Problem Statement

Phishing attacks are one of the most prevalent cybersecurity threats, targeting users through deceptive URLs that impersonate legitimate websites to steal credentials and financial data. Traditional rule-based detection systems struggle to keep up with evolving phishing techniques such as homograph attacks, URL encoding, and short URL obfuscation. This project builds an end-to-end intelligent phishing URL detection system that combines structured feature engineering, deep learning on raw URL characters, explainable AI, and real-time threat intelligence.

2. Methodology

Feature Extraction Pipeline

A three-layer feature extraction pipeline was built to extract 26 features from any raw URL: (1) Lexical features — URL length, entropy, special characters, HTTPS presence, IP-based URLs; (2) WHOIS features — domain age, expiration days, registrar availability; (3) DNS features — A record, MX record, NS record existence and count.

Model Architecture

Two models were trained and combined into a weighted ensemble. A Gradient Boosting Classifier (GBC) was trained on 87 structured features from 11,430 labeled URLs. A character-level LSTM was trained on raw URL strings tokenized at the character level, with a vocabulary of 101 unique characters and sequences padded to length 200. The ensemble combines GBC (60% weight) and LSTM (40% weight) predictions into a final risk score.

Explainability & Threat Intelligence

SHAP (SHapley Additive exPlanations) TreeExplainer was applied to the GBC model to generate per-prediction feature attributions. Each URL analysis returns the top 5 features driving the classification decision. VirusTotal API integration cross-checks URLs against 70+ antivirus engines, contributing 20% weight to the final risk score when available.

3. Datasets

Dataset	Size	Purpose
Kaggle Phishing Dataset	11,054 URLs	GBC training (pre-extracted features)
Web Page Phishing Detection	11,430 URLs	LSTM + GBC training (raw URLs)
PhishTank verified_online.csv	58,489 URLs	Additional phishing URL source

4. Results

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
GBC	96.47%	96.23%	97.48%	96.85%	99.47%
LSTM	89.37%	88.73%	90.20%	89.46%	95.70%
Ensemble	96.76%	96.44%	97.11%	96.77%	99.18%

The ensemble model achieves the highest accuracy (96.76%) and F1 score (96.77%), outperforming both individual models. The GBC AUC-ROC of 99.47% indicates near-perfect class separation on structured features. The LSTM achieves

89.37% accuracy using only raw URL characters with no additional metadata, demonstrating strong pattern recognition from character sequences alone.

5. Adversarial Testing

Attack Type	Example	Detected?	Confidence
URL Shortener	bit.ly/3xampleURL	Yes	99.74%
TinyURL	tinyurl.com/y4example	Yes	97.34%
URL Encoding	paypal%2Ecom/login	Yes	97.84%
Homograph (l vs I)	paypal.com/login	Yes	78.76%
Subdomain Trick	google.com.phishing-site.com	Yes	99.81%
HTTPS Phishing	secure-paypal-login.com	Yes	99.75%
Encoded Dots	www%2Egoogle%2Ecom	No	1.64%
Subtle Homograph	arnazon.com (rn vs m)	No	26.60%

6. Limitations & Future Work

Limitations: The system fails to detect URL-encoded domain attacks (e.g., www%2Egoogle%2Ecom) because percent-encoding makes the URL appear short and clean to the character-level model. Subtle homograph attacks (e.g., arnazon.com) also evade detection as the model lacks brand awareness. The GBC model relies on pre-extracted features and falls back to default values for live URLs not present in the training dataset. VirusTotal integration is limited to 500 requests/day on the free tier.

Future Work: Implement URL decoding as a preprocessing step before feature extraction. Add brand similarity checking using edit distance against a curated brand list. Train on a larger dataset (500k+ URLs) to improve LSTM generalization. Deploy the API to cloud infrastructure (AWS/GCP) for public access. Develop a browser extension for real-time phishing detection.

7. Tech Stack

Component	Tool/Library
Language	Python 3.10
ML Models	scikit-learn (GBC), TensorFlow/Keras (LSTM)
Explainability	SHAP (TreeExplainer)
URL Features	python-whois, dnspython, tldextract
Threat Intelligence	VirusTotal API (free tier)
API Framework	FastAPI + Uvicorn
Data Processing	pandas, numpy
Visualization	matplotlib, seaborn
Version Control	Git + GitHub